



Group I
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INTRODUCTION

Demand is growing +80% per year but the logistics network hasn't changed since 2019, costs are about to outpace revenues.

- Founded 2014, **e-commerce sports equipment** for families
- Mix of **new** and **second-hand products**, selling price ~**\$14/unit**
- Currently: **1 small warehouse in Saint-Louis, serving all of the US**
- **All suppliers ship to Saint-Louis, all customers served from Saint-Louis via UPS (48h)**

2024 → 2027

×5.8

demand growth in 3 years

4M units

Saint Louis Max Capacity

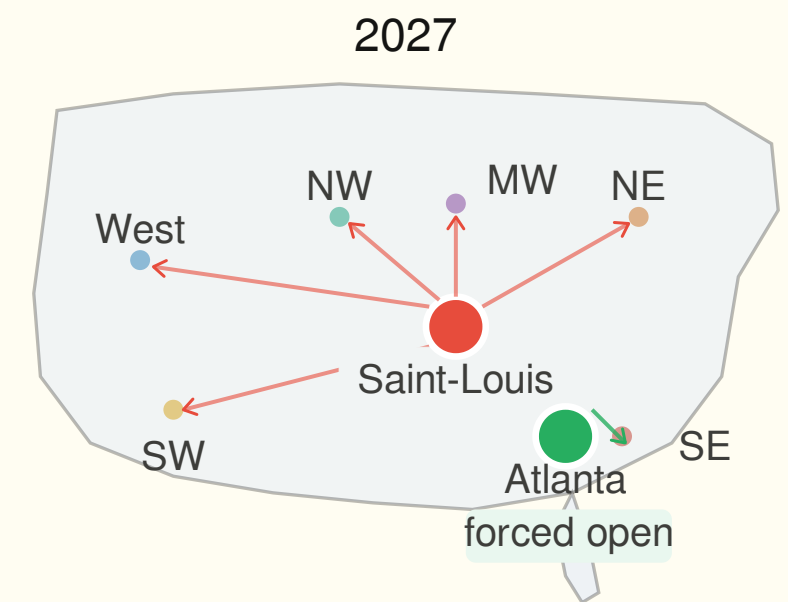
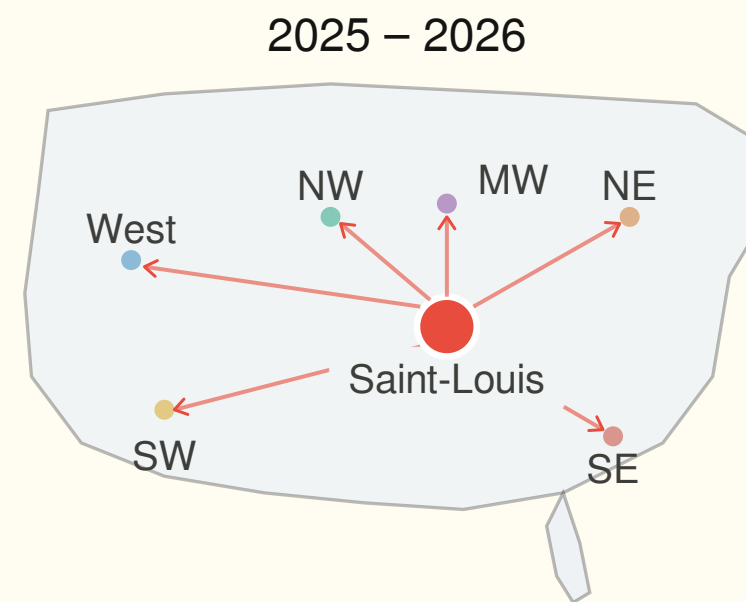
8.3M units

Demand by 2027



SAINT LOUIS ONLY

- **2025 & 2026** → Saint-Louis Large handles everything, but ships inefficiently to the West Coast
- **2027** → Saint-Louis hits its 4M unit capacity limit: Atlanta must open anyway
- **Geography ignored** → shipping West from Saint-Louis costs \$3/shipment vs \$2.50 from Denver



● Saint-Louis ● Atlanta (forced)

→ *Saint-Louis can handle half of what customers will need by 2027*



BASELINE



2025: Saint Louis Large Only

- Serves **all 6 regions from Saint-Louis**
- Same as today but upgraded to Large
 - 2M to 4M
- Makes sense: **demand is manageable at 2.6M units, STL Large fits**

2026: Saint-Louis Large + Denver Small

- Denver takes over West and Southwest regions
- Why Denver? **Cheapest UPS rate to West (\$2.50) vs Saint-Louis (\$3.00)**
- Saint-Louis handles Northwest, Midwest, Northeast, Southeast

2027: Saint-Louis Large + Denver Large + Atlanta Small

- **Demand** explodes to **8.3M** → **need 3 warehouses**
- Denver upgrades to **Large**, takes **West, Southwest, part of Northwest**
- Atlanta opens **Small**, takes **Southeast**
- **Saint-Louis** handles **Midwest, Northeast, rest of Northwest**

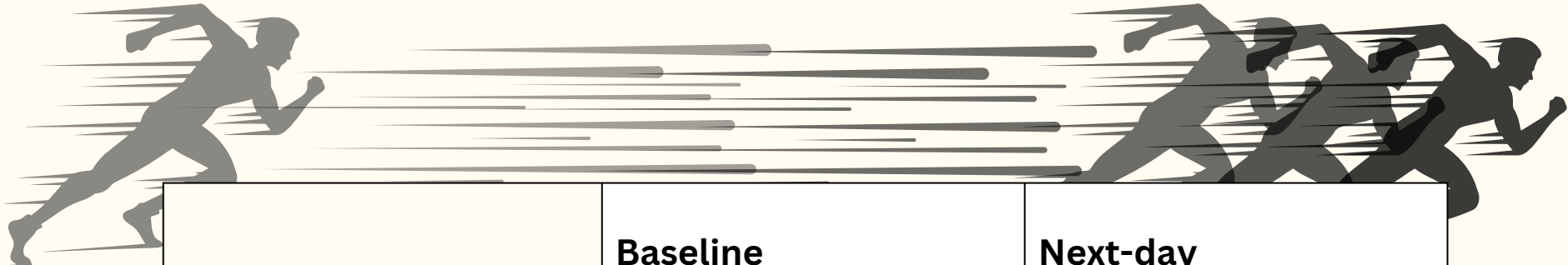
The optimal solution expands the network progressively:

- **1 warehouse in 2025**
- **2 in 2026**
- **3 in 2027**

→ *saving costs by getting closer to customers.*

	Q1 Saint-Louis only	Q2 Optimal
3-Year Total	Not feasible	\$187,034,219

NEXT-DAY DELIVERY



- *Optimal solution keeps the same network structure as the Baseline model.*
- *Total costs increase in 2027 due to higher outbound transportation costs and the +10% demand uplift, confirming that purchasing cost remains the main cost driver (~82% of total costs).*

	Baseline	Next-day
3-Year Total cost	187,034,219.00	197,872,158.00
Delta Sales	11,634,840.00	
Gain	796,901.00	

2025: Saint Louis Large Only

Same network as Baseline: next-day delivery **not yet active**, no additional cost or demand impact

2026: Saint-Louis Large + Denever Small

Same network as Baseline: next-day delivery **still not active**, **DEN Small opens naturally** due to +80% demand growth

2027: Saint-Louis Large + Denver Large + Atlanta Small

Next-day delivery launches → demand +10% in all regions, **UPS costs +20%** → **3 warehouses** required to absorb 9.1M units. Net gain of only \$796K despite \$10.8M in additional purchasing costs

RESILIENT



SportB operates from one small warehouse in Saint-Louis, rated "**Relatively High**" on the FEMA Risk Map. With 4 weeks of disruption per year historically, a single closure in 2027 leaves the entire network exposed.

2025: Saint Louis Small +
Atlanta Small

2026: Denver Small + Saint
Louis Large

2027: Saint-Louis Large + Denver
Large + Atlanta Small

Worst case: Saint-Louis close in 2027

Units to redirect

307,692

7.7% of annual flow

Backup capacity available

5,668,415

Shock fully absorbed

Revenue protected

\$4.3M

In 4 weeks of closure

3-year premium cost

+\$561K

Spending **\$561,875** over 3 years protects up to **\$9.6M** in revenue across all disruption scenarios

Baseline cost \$187,034,219

Resilient cost \$187,596,094

IN-SOURCING



→ Instead of buying refurbished products at \$10/unit, SportB does it in-house at \$8/unit, but needs to double its capital investment.

The saving:

- Current cost: \$10/unit
- In-sourced cost: \$8/unit (80% of products)
- New products stay at: \$10/unit (20%)
- Blended new cost: \$8.40/unit
- 3-year saving on purchases: \$24.8M

The cost:

- Requires doubling capital employed
- Extra capital per year: \$14.4M
- Cost of capital (WACC 10%): \$4.3M over 3 years
- Net benefit: +\$20.5M

- SportB currently buys all products from US suppliers at \$10/unit. By refurbishing second-hand products in-house (80% of sales), **the blended cost drops to \$8.40/unit, saving \$24.8M over 3 years.**
- However, in-sourcing requires doubling the capital employed. Based on an estimated capital base of **40% of revenue**, this means **\$14.4M** in extra capital per year. At a **WACC of 10%**, the cost of that capital is **\$4.3M over 3 years.**
- **Net benefit: +\$20.5M** ✓

Assumptions: capital base = 40% of annual revenue · WACC = 10%

	Q2 Baseline	Q6 In-sourcing	Saving
3-Year Total Cost	\$187'034'219	\$162'246'059	-\$24.8M

INTERNATIONAL

Same as the baseline model :

2025: Saint Louis Large Only

2026: Saint-Louis Large + Denver Small

2027: Saint-Louis Large + Denver Large + Atlanta Small

	Baseline Model	International Model
Total cost	\$187,034,219	\$182,868,249
Total Profit	\$29,862,181	\$34,028,151

Key takeaway: ~\$4.2M in net savings over 3 years vs. Baseline

Drastic Inbound Logistics Optimization:

- **-\$4.5M reduction** in inbound transport costs
- **Strategic Sourcing Shift:**
 - e.g. leveraged Juarez (Mexico) to supply the Seattle warehouse → **Cut per-shipment cost by more than half to \$1.50** (vs. \$3.50 US baseline)

Profitable Strategic Trade-Off :

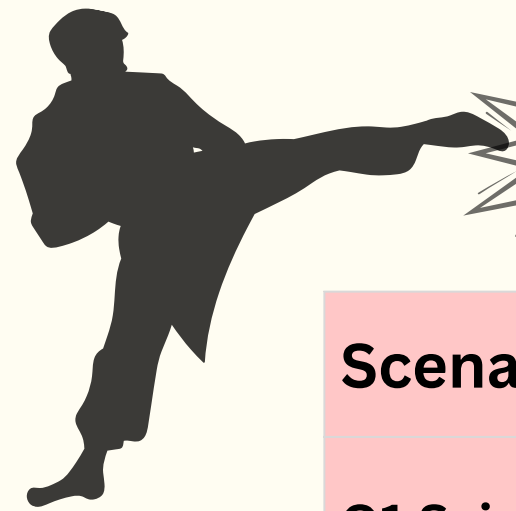
- **Calculated Cost Increase:** Absorbed **+\$772K** in fixed warehouse fees and **+\$1.0M** in inventory holding costs.
- **Net Positive ROI:** This structural investment is heavily offset by accessing highly competitive international freight and purchase prices.

→ **Mitigation of single-source domestic dependency**



The Trade-Off: This strategy secures significant cost reductions, but takes on major exposure linked to geopolitical risks and longer international supply chains that could generate important costs.

Possible Solution: Implement strategic inventory buffers and dual-sourcing for critical components to protect our margins from sudden disruption costs.



CONCLUSION

Scenario	Recommendation	Key number	Total Cost	Profit
Q1 Saint-Louis only	Not recommended	Capacity breaks in 2027	/	/
Q2 Optimal network	Do this now	Cheapest 3-year solution	187,034,219.00	29,862,181.00
Q3 Next-day delivery	⚠ Evaluate in 2026	Check delta revenue vs cost	197,872,158.00	30,659,082.00
Q4 Resilient network	Worth it	\$453K premium, 17x ROI	187,596,094.00	29,300,306.00
Q5 International sourcing	⚠ Consider from 2026	Cheaper but geopolitical risk	182,868,249.00	34,028,151.00
Q6 In-sourcing	Plan for post-2027	+\$20.5M but needs expertise	162,246,059.00	54,650,341.00



- 2025: Operate Saint-Louis (S) and Atlanta (S) to secure demand coverage while accepting a limited cost premium to improve network resilience
- 2026: Upgrade Saint-Louis to Large and Denver (S) to further stabilize the network and prepare for future growth with controlled additional costs
- 2027: Activate international sourcing through Juarez and Suzhou if geopolitical conditions remain stable → -\$3M inbound savings in 2027
- Post-2027: Launch in-sourcing once the network is stabilized and after further capital cost assessment, reducing purchase costs by \$24.8M over 3 years → Finally transform the operating model

***THANK YOU FOR
LISTENING***



APPENDIX



ST LOUIS ONLY

Constraints:

- C1: Demand Satisfaction
 $\sum w x(w,r,t) = \text{demand}(r,t)$
- C2: Warehouse Capacity
 $\sum r x(w,r,t) \times 4 \leq \text{capacity}(w,t)$
- C3: One Size per Warehouse
 $y(w,S,t) + y(w,L,t) \leq 1$
- C4: Opening / Closing Transfer Logic
 $z(w,s,t) \geq y(w,s,t-1) - y(w,s,t)$
- C5: Supplier Capacity
 $\sum w \text{FluxIn}(US,w,t) \leq 17,000,000$
- C6: Flow Balance
 $\text{FluxIn}(w,t) = \sum r x(w,r,t)$
- C7: STL Large Only
 $y(STL,L,t) = 1$
- C8 Other Warehouses Closed
 $y(w,s,t) = 0$
- Binary Constraint
 $y(w,s,t) \in \{0,1\}$

Cost Component	2025	2026	2027	3-Year Total
1. Warehouse Fixed Costs (\$/yr)	\$375 000	\$467 550	\$849 166	\$1 691 716
2. Warehouse Variable Costs (WH handling)	\$513 000	\$923 400	\$1 662 120	\$3 098 520
3. Warehouse Holding Costs	\$898 225	\$1 383 343	\$2 870 017	\$5 151 584
4. UPS Outbound (WH → customers)	\$1 877 625	\$3 225 475	\$5 828 355	\$10 931 455
5. UPS Inbound (Supplier → WH) (US SOURCE)	\$1 603 125	\$3 039 875	\$5 732 950	\$10 375 950
6. Product Purchase Cost (US sources)	\$25 650 000	\$46 170 000	\$83 106 000	\$154 926 000
7. Warehouse Transfer/Close Costs	30000	0	9255	\$39 255
8. Fixed costs supplier	\$20 000	\$20 000	\$20 000	\$60 000
9. Variable costs supplier	192375	346275	623295	\$1 161 945
TOTAL COST (Objective)	\$31 159 350	\$55 575 918	\$100 701 158	

Assumption

1. Network structure Saint-Louis is the unique warehouse throughout the entire 2025–2027 planning

BASE LINE

Constraints:

- C1: Demand Satisfaction
 $\sum w x(w,r,t) = \text{demand}(r,t)$
- C2: Warehouse Capacity
 $\sum r x(w,r,t) \times 4 \leq \text{capacity}(w,t)$
- C3: One Size per Warehouse
 $y(w,S,t) + y(w,L,t) \leq 1$
- C4: Opening / Closing Transfer Logic
 $z(w,s,t) \geq y(w,s,t-1) - y(w,s,t)$
- C5: Supplier Capacity
 $\sum w \text{FluxIn}(US,w,t) \leq 17,000,000$
- C6: Flow Balance
 $\text{FluxIn}(w,t) = \sum r x(w,r,t)$
- Binary Constraint
 $y(w,s,t) \in \{0,1\}$

Assumption

- Each order corresponds to 4 units.
- No backlog or lost sales allowed: all demand must be satisfied.
- Opening, closing, or resizing a warehouse generates a fixed transfer cost of \$30,000.
- The model is deterministic and solved as a Mixed Integer Program (MIP) using OpenSolver / CBC.

Cost Component	2025	2026	2027	3-Year Total
1. Warehouse Fixed Costs (\$/yr)	\$375 000	\$625 000	\$1 015 000	\$2 015 000
2. Warehouse Variable Costs (WH handling)	\$513 000	\$923 400	\$1 662 120	\$3 098 520
3. Warehouse Holding Costs	\$898 225	\$1 711 805	\$2 796 249	\$5 406 279
4. UPS Outbound (WH → customers)	\$1 877 625	\$3 169 125	\$5 449 275	\$10 496 025
5. UPS Inbound (Supplier → WH) (US SOURCE)	\$1 603 125	\$2 885 625	\$5 321 700	\$9 810 450
6. Product Purchase Cost (US sources)	\$25 650 000	\$46 170 000	\$83 106 000	\$154 926 000
7. Warehouse Transfer/Close Costs	30000	0	30000	\$60 000
8. Fixed costs supplier	\$20 000	\$20 000	\$20 000	\$60 000
9. Variable costs supplier	192375	346275	623295	\$1 161 945
TOTAL COST (Objective)	\$31 159 350	\$55 851 230	\$100 023 639	\$187 034 219
TOTAL REVENU (14\$*unit)	\$ 35 910 000	\$ 64 638 000	\$ 116 348 400	\$ 216 896 400
PROFIT	\$ 4 750 650	\$ 8 786 770	\$ 16 324 761	\$ 29 862 181

FAST DELIVERY

Constraints:

- C1: Demand Satisfaction**
 $\sum w x(w,r,t) = \text{demand}(r,t)$
- C2: Warehouse Capacity**
 $\sum r x(w,r,t) \times 4 \leq \text{capacity}(w,t)$
- C3: One Size per Warehouse**
 $y(w,S,t) + y(w,L,t) \leq 1$
- C4: Opening / Closing Transfer Logic**
 $z(w,s,t) \geq y(w,s,t-1) - y(w,s,t)$
- C5: Supplier Capacity**
 $\sum w \text{FluxIn}(US,w,t) \leq 17,000,000$
- C6: Flow Balance**
 $\text{FluxIn}(w,t) = \sum r x(w,r,t)$
- Binary Constraint**
 $y(w,s,t) \in \{0,1\}$

Assumption

- 2027 demand increased by +10%
- UPS outbound transportation costs increased by +20%
- 2025–2026 parameters unchanged vs Baseline
- Next-day delivery option introduced in 2027 only (2025–2026 unchanged)

Cost Component	2025	2026	2027	3-Year Total
1. Warehouse Fixed Costs (\$/yr)	\$375 000	\$625 000	\$1 015 000	\$2 015 000
2. Warehouse Variable Costs (WH handling)	\$513 000	\$923 400	\$1 828 332	\$3 264 732
3. Warehouse Holding Costs	\$898 225	\$1 711 805	\$2 933 374	\$5 543 404
4. UPS Outbound (WH → customers)	\$1 877 625	\$3 169 125	\$7 193 043	\$12 239 793
5. UPS Inbound (Supplier → WH) (US SOURCE)	\$1 603 125	\$2 885 625	\$5 739 605	\$10 228 355
6. Product Purchase Cost (US sources)	\$25 650 000	\$46 170 000	\$91 416 600	\$163 236 600
7. Warehouse Transfer/Close Costs	30000	0		\$60 000
8. Fixed costs supplier	\$20 000	\$20 000	\$20 000	\$60 000
9. Variable costs supplier	192375	346275	6,856,245	\$1 224 275
TOTAL COST (Objective)	\$31 159 350	\$55 851 230	\$110 861 578	\$197 872 158
TOTAL REVENU (14\$*unit)	\$ 35 910 000	\$ 64 638 000	\$127'983'240	\$ 228'531'240
PROFIT	\$ 4 750 650	\$ 8 786 770	\$17'121'662	\$30'659'082

RISK RESILIENT

Constraints:

- C1: Demand Satisfaction
 $\sum w x(w,r,t) = \text{demand}(r,t)$
- C2: Warehouse Capacity
 $\sum r x(w,r,t) \times 4 \leq \text{capacity}(w,t)$
- C3: One Size per Warehouse
 $y(w,S,t) + y(w,L,t) \leq 1$
- C4: Opening / Closing Transfer Logic
 $z(w,s,t) \geq y(w,s,t-1) - y(w,s,t)$
- C5: Supplier Capacity
 $\sum w \text{FluxIn}(US,w,t) \leq 17,000,000$
- C6: Flow Balance
 $\text{FluxIn}(w,t) = \sum r x(w,r,t)$
- Binary Constraint
 $y(w,s,t) \in \{0,1\}$

Assumption

- Each order corresponds to 4 units.
- No backlog or lost sales allowed: all demand must be satisfied.
- Opening, closing, or resizing a warehouse generates a fixed transfer cost of \$30,000.
- The model is deterministic and solved as a Mixed Integer Program (MIP) using OpenSolver / CBC.

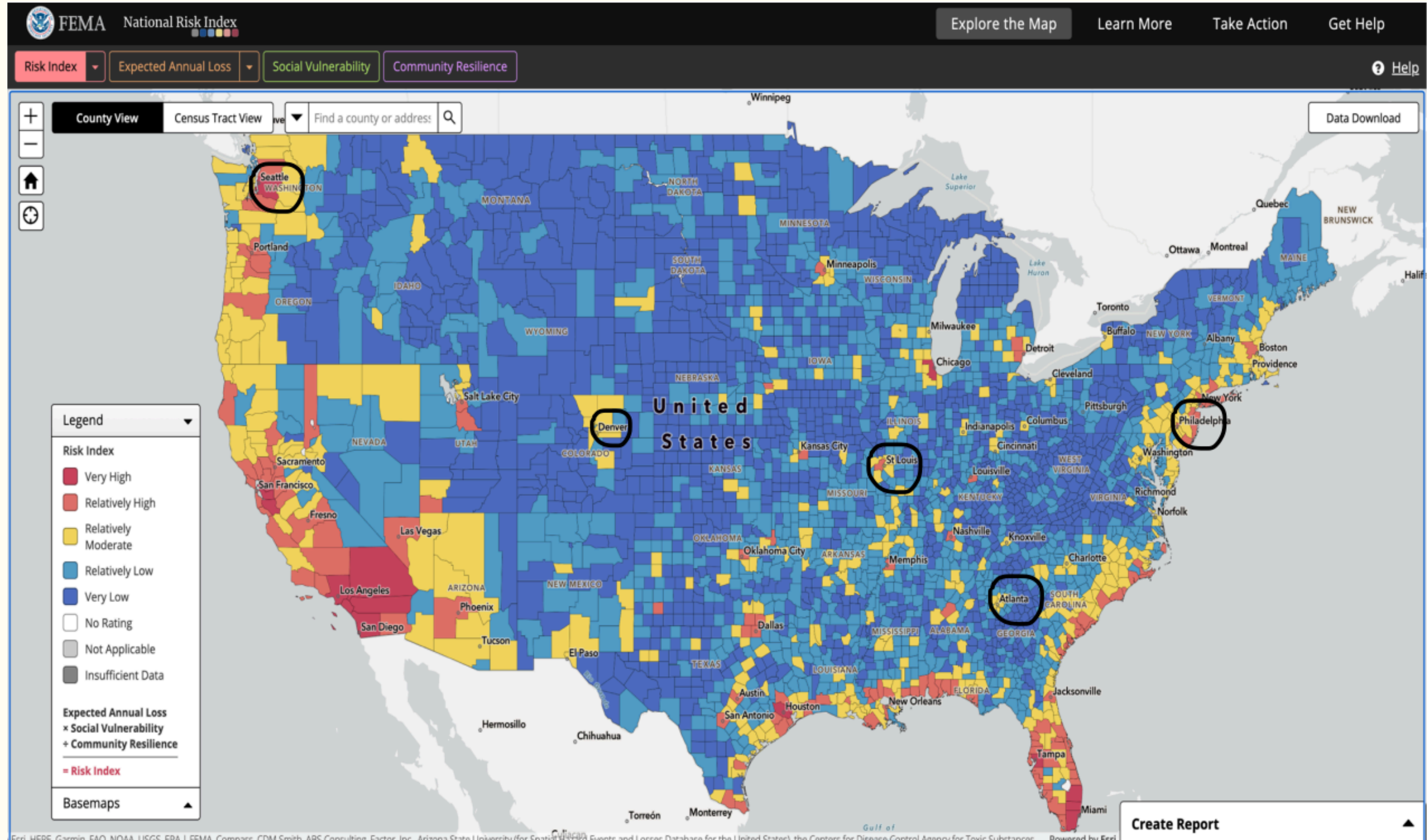
Cost Component	2025	2026	2027	3-Year Total
1. Warehouse Fixed Costs (\$/yr)	\$440 000	\$625 000	\$1 015 000	\$2 080 000
2. Warehouse Variable Costs (WH handling)	\$513 000	\$923 400	\$1 662 120	\$3 098 520
3. Warehouse Holding Costs	\$1 373 225	\$1 711 805	\$2 796 249	\$5 881 279
4. UPS Outbound (WH → customers)	\$1 798 875	\$3 169 125	\$5 449 275	\$10 417 275
5. UPS Inbound (Supplier → WH) (US SOURCE)	\$1 673 750	\$2 885 625	\$5 321 700	\$9 881 075
6. Product Purchase Cost (US sources)	\$25 650 000	\$46 170 000	\$83 106 000	\$154 926 000
7. Warehouse Transfer/Close Costs	0	60000	30000	\$90 000
8. Fixed costs supplier	\$20 000	\$20 000	\$20 000	\$60 000
9. Variable costs supplier	192375	346275		\$1 161 945
TOTAL COST (Objective)	\$31 661 225	\$55 911 230	\$100 023 639	\$187 596 094
TOTAL REVENU (14\$*unit)	\$ 35 910 000	\$ 64 638 000	\$ 116 348 400	\$ 216 896 400
PROFIT	\$ 4 248 775	\$ 8 726 770	\$ 16 324 761	\$ 29 300 306

RISK RESILIENT

RISK ANALYSIS				
	2025	2026	2027	3-years
Resilient network cost	\$31'661'225	\$55'911'2	\$100'023'	\$187'596'09
Baseline network cost	\$31'159'350	\$55'851'2	\$100'023'	\$187'034'21
RESILIENCE PREMIUM				\$561'875

DISRUPTION SCENARIO					
		2025	2026	2027	TOTAL 3 YEARS
STL	annual flow (units)	2'000'000	2'932'200	4'000'000	8'932'200
	capacity (units)	2'000'000	4'000'000	4'000'000	10'000'000
	flow at risk (units)	153'846	225'554	307'692	
	available capacity	1'846'154	3'774'446	3'692'308	
DEN	annual flow (units)	0	1'684'800	3'290'000	4'974'800
	capacity (units)	0	2'000'000	4'000'000	6'000'000
	flow at risk (units)	0	129'600	253'077	382'677
	available capacity	0	1'870'400	3'746'923	5'617'323
ATL	annual flow (units)	565000	0	1020600	1'585'600
	capacity (units)	2'000'000	0	2'000'000	4'000'000
	flow at risk (units)	43'462	0	78'508	121'969
	available capacity	1'956'538	0	1'921'492	3'878'031
RESILIENCE CHECK					
STL Closing	Units to redirect	153'846	225'554	307'692	
	Total backup capacity	1'956'538	1'870'400	5'668'415	
	Margin (backup - redirect)	1'802'692	1'644'846	5'360'723	
	Revenue protected (\$14/unit)	\$2'153'846	\$3'157'754	\$4'307'692	\$9'619'292
DEN Closing	Units to redirect	0	129'600	253'077	
	Total backup capacity	0	3'774'446	5'613'800	
	Margin (backup - redirect)	0	3'644'846	5'360'723	
	Revenue protected (\$14/unit)	\$0	\$1'814'400	\$3'543'077	\$5'357'477
ATL Closing	Units to redirect	43'462	0	78'508	
	Total backup capacity	1'846'154	0	7'439'231	
	Margin (backup - redirect)	1'802'692	0	7'360'723	
	Revenue protected (\$14/unit)	\$608'462	\$0	\$1'099'108	\$1'707'569

RISK RESILIENT



INTERNATIONAL

Constraints:

- C1: Demand Satisfaction
 $\sum w x(w,r,t) = \text{demand}(r,t)$
- C2: Warehouse Capacity
 $\sum r x(w,r,t) \times 4 \leq \text{capacity}(w,t)$
- C3: One Size per Warehouse
 $y(w,S,t) + y(w,L,t) \leq 1$
- C4: Opening / Closing Transfer Logic
 $z(w,s,t) \geq y(w,s,t-1) - y(w,s,t)$
- C5: Supplier Capacity
 $\sum w \text{FluxIn}(k,w,t) \leq \text{CAP}_k \times u(k,t) \quad \forall k \in \{\text{Suzhou, Juarez, US}\}$
- C6: Flow Balance
 $\sum k \text{FluxIn}(k,w,t) = \sum r x(w,r,t)$
- Binary Constraint
 $y(w,s,t) \in \{0,1\}$
 $u(k,t) \in \{0,1\} \quad \forall k \in \{\text{Suzhou, Juarez, US}\}$

Assumption

- Three sourcing options available: Suzhou (China, cap. 3M units), Juarez (Mexico, cap. 6M units), US Sources (cap. 17M units)
- A supplier is contracted only if $u(k,t) = 1$, which activates its fixed annual contract cost

Cost Component	2025	2026	2027	3-Year Total
1. Warehouse Fixed Costs (\$/yr)	\$500 000	\$720 000	\$1 095 000	\$2 315 000
2. Warehouse Variable Costs (WH handling)	\$513 000	\$923 400	\$1 662 120	\$3 098 520
3. Warehouse Holding Costs	\$898 225	\$1 711 805	\$2 796 249	\$5 406 279
4. UPS Outbound (WH → customers)	\$2 381 625	\$3 324 300	\$5 215 995	\$10 921 920
5. UPS Inbound (Supplier → WH)	\$961 875	\$2 231 375	\$2 077 650	\$5 270 900
6. Product Purchase Cost	\$25 650 000	\$46 170 000	\$83 106 000	\$154 926 000
7. Warehouse Transfer/Close Costs	30000	0	30000	\$60 000
8. Fixed costs supplier	30000	50000	40000	\$120 000
9. Variable costs supplier	128250	280850	340530	\$749 630
TOTAL COST (Objective)	\$31 092 975	\$55 411 730	\$96 363 544	\$182 868 249
TOTAL REVENU (14\$*unit)	\$ 35 910 000	\$ 64 638 000	\$ 116 348 400	\$ 216 896 400
PROFIT	\$ 4 817 025	\$ 9 226 270	\$ 19 984 856	\$ 34 028 151

INSOURCING

Constraints:

- C1: Demand Satisfaction
 $\sum w \cdot x(w,r,t) = \text{demand}(r,t)$
- C2: Warehouse Capacity
 $\sum r \cdot x(w,r,t) \times 4 \leq \text{capacity}(w,t)$
- C3: One Size per Warehouse
 $y(w,S,t) + y(w,L,t) \leq 1$
- C4: Opening / Closing Transfer Logic
 $z(w,s,t) \geq y(w,s,t-1) - y(w,s,t)$
- C5: Supplier Capacity
 $\sum w \cdot \text{FluxIn}(k,w,t) \leq \text{CAP}_k \times u(k,t) \quad \forall k \in \{\text{Suzhou, Juarez, US}\}$
- C6: Flow Balance
 $\sum k \cdot \text{FluxIn}(k,w,t) = \sum r \cdot x(w,r,t)$
- Binary Constraint
 $y(w,s,t) \in \{0,1\}$
 $u(k,t) \in \{0,1\} \quad \forall k \in \{\text{Suzhou, Juarez, US}\}$

Assumption

- Three sourcing options available: Suzhou (China, cap. 3M units), Juarez (Mexico, cap. 6M units), US Sources (cap. 17M units)
- A supplier is contracted only if $u(k,t) = 1$, which activates its fixed annual contract cost

sensitive analysis				
	2025	2026	2027	total
capital baseline	\$ 14 364 000	\$ 14 364 000	14 364 000	\$ 14 364 000
capital	\$ 28 728 000	\$ 28 728 000	28 728 000	\$ 28 728 000
capital	\$ 14 364 000	\$ 14 364 000	14 364 000	\$ 14 364 000
WACC 10%	\$ 1 436 400	\$ 1 436 400,0	1 436 400	\$ 1 436 400
Economic	\$ 4 104 000	\$7 387 200	13 296 960	\$24 788 160
si rentable in	\$ 2 667 600	\$ 5 950 800	\$ 11 860 560	\$ 23 351 760

SECTION 3 – OBJECTIVE FUNCTION				
Cost Component	2025	2026	2027	3-Year Total
1. Warehouse Fixed Costs (\$/yr)	\$375 000	\$625 000	\$1 015 000	\$2 015 000
2. Warehouse Variable Costs (WH handling)	\$513 000	\$923 400	\$1 662 120	\$3 098 520
3. Warehouse Holding Costs	\$898 225	\$1 711 805	\$2 796 249	\$5 406 279
4. UPS Outbound (WH → customers)	\$1 877 625	\$3 169 125	\$5 449 275	\$10 496 025
5. UPS Inbound (Supplier → WH) (US SOURCE)	\$1 603 125	\$2 885 625	\$5 321 700	\$9 810 450
6. Product Purchase Cost (US sources)	\$21 546 000	\$38 782 800	\$69 809 040	\$130 137 840
7. Warehouse Transfer/Close Costs	30000	0	30000	\$60 000
8. Fixed costs supplier	\$20 000	\$20 000	\$20 000	\$60 000
9. Variable costs supplier	192375	346275	623295	\$1 161 945
TOTAL COST (Objective)	\$27 055 350	\$48 464 030	\$86 726 679	\$162 246 059
TOTAL REVENU (14\$*unit)	\$ 35 910 000	\$ 64 638 000	\$ 116 348 400	\$ 216 896 400
PROFIT	\$ 8 854 650	\$ 16 173 970	\$ 29 621 721	\$ 54 650 341

Comparison	baseline	insourcing	Economic purchase
total	\$ 187 034 219	\$ 162 246 059	\$ 24 788 160
2025	\$ 31 159 350	\$ 27 055 350	\$ 4 104 000
2026	\$ 55 851 230	\$ 48 464 030	\$ 7 387 200
2027	\$ 100 023 639	\$ 86 726 679	\$ 13 296 960